

Thesis/Capstone Archiving and Host Training Establishment (HTE)
For CCS Graduating Students

A Capstone Project presented to the
College of Computer Studies of
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

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TABLE OF CONTENTS



ABSTRACT

Title : Thesis/Capstone Archiving and Host Training
Establishment (HTE) For CCS Graduating Students

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Degree : Bachelor of Science in Information Technology

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Project Description : This study presents the development of a web-based system designed to improve the management of thesis/capstone projects and Host Training Establishment (HTE) coordination for graduating students of the College of Computer Studies (CCS). The system addresses common issues such as inefficient archiving, difficulty in retrieving past research outputs, and the lack of a centralized platform for monitoring student deployment in HTEs. The proposed system allows administrators, coordinators, and students to manage, store, and access thesis documents digitally while also facilitating the tracking of student assignments, progress,



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College of Computer Studies



and performance in their respective training establishments. It features role-based access, search and filtering capabilities, report generation, and secure data storage.

The system was developed using an Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Results from user testing indicated that the system is effective, user-friendly, and capable of significantly improving the efficiency of thesis archiving and HTE management processes. The implementation of this system provides a centralized, organized, and accessible solution that benefits students, faculty, and administrators, and serves as a foundation for future enhancements such as mobile integration and advanced analytics.



CHAPTER 1: INTRODUCTION

I. Introduction

In academic institutions, particularly within the College of Computer Studies (CCS), managing thesis and capstone projects has become increasingly complex due to the growing number of students and outputs each year. Traditional archiving methods, such as physical storage and scattered digital files, often result in inefficiency, data loss, and difficulty in retrieval.

Additionally, the lack of a centralized platform for Host Training Establishment (HTE) coordination creates challenges in tracking student deployment, monitoring progress, and ensuring proper documentation.

This study proposes a web-based system designed to streamline thesis archiving and improve HTE coordination for graduating students.

II. Significance of the Study

This system will benefit the following:

- **Students** – Easy access to previous capstones and HTE tracking
- **Faculty** – Simplified evaluation and monitoring
- **Administrators** – Centralized data management
- **Future Researchers** – Accessible reference materials



III. Statement of the Problem

General Problem

How can a centralized system improve thesis archiving and HTE management for CCS graduating students?

Specific Problems

1. How to store and retrieve capstone projects efficiently?
2. How to monitor student HTE assignments and progress?
3. How to ensure data security and accessibility?
4. How to generate reports for academic and administrative use?

IV. Scope and Limitations

Scope

- Archiving of thesis/capstone projects
- HTE management (deployment, tracking)
- User roles: Admin, Coordinator, Student

Limitations

- Limited to CCS department only
- Requires internet connection
- No AI-based recommendations (for now)



OPERATIONAL DEFINITION OF TERMS

Administrator – Refers to the authorized personnel responsible for managing the system, including user accounts, thesis archives, and system configurations.

Application Programming Interface (API) – A set of protocols that allows different software components of the system to communicate and exchange data.

Archiving – The process of storing, organizing, and maintaining digital thesis or capstone files for long-term access and retrieval.

Authentication – The process of verifying a user's identity through login credentials before granting system access.

Authorization – The process of determining user permissions and access levels based on assigned roles.

Backend – The server-side component of the system responsible for processing logic, handling requests, and managing database operations.

Capstone Project – A final academic requirement completed by CCS students, stored and managed within the system.

Cloud Storage – An online storage system that allows data to be stored and accessed remotely via the internet.

College of Computer Studies (CCS) – The academic department where the system is implemented and utilized.



Coordinator – A faculty member responsible for overseeing student deployment and monitoring HTE activities using the system.

Data Backup – The process of creating duplicate copies of data to prevent loss in case of system failure.

Data Security – Measures implemented to protect system data from unauthorized access, breaches, or loss.

Database – A structured collection of data that stores user information, thesis records, and HTE details.

Database Management System (DBMS) – Software used to create, manage, and manipulate the database efficiently.

Document Repository – A centralized storage module where thesis and capstone documents are uploaded and accessed.

Encryption – A method of converting sensitive information into a secure format to prevent unauthorized access.

File Upload Module – A feature that enables users to upload thesis documents and related files into the system.

Frontend – The client-side interface where users interact with the system through web pages.

Functional Suitability (ISO 25010) – The degree to which the system provides functions that meet stated and implied needs when used under specified conditions.



Hosting Server – The environment where the web application is deployed and accessed over a network.

Host Training Establishment (HTE) – The organization or company where students undergo training or internship.

HTE Monitoring – The process of tracking student activities, attendance, and performance in their assigned HTE.

Hypertext Transfer Protocol Secure (HTTPS) – A secure protocol used for transmitting data between the user and the system.

Indexing – The process of organizing data to improve the speed and efficiency of search operations.

Input Validation – The process of checking and ensuring that user inputs are accurate, valid, and secure before processing.

Login Session – A temporary session created after successful authentication, allowing user interaction with the system.

Maintainability (ISO 25010) – The ease with which the system can be modified to correct faults, improve performance, or adapt to changes.

Modular Design – A design approach where the system is divided into separate modules to improve maintainability and scalability.

Object-Oriented Programming (OOP) – A programming paradigm that structures the system using objects and classes.



Performance Efficiency (ISO 25010) – The system’s ability to perform tasks efficiently with optimal use of resources and response time.

Portability (ISO 25010) – The ease with which the system can be transferred and used across different environments or platforms.

Relational Database – A type of database that organizes data into tables with relationships between them.

Reliability (ISO 25010) – The ability of the system to perform consistently and accurately under specified conditions over time.

Report Generation – A system feature that produces organized outputs and summaries based on stored data.

Role-Based Access Control (RBAC) – A security mechanism that restricts system access based on user roles.

Search Algorithm – A method used by the system to locate specific records efficiently based on user queries.

Search and Retrieval – A system function that allows users to find archived documents using filters and keywords.

Security (ISO 25010) – The system’s ability to protect information and data from unauthorized access and ensure confidentiality, integrity, and availability.

System Integration – The process of combining multiple modules into a unified system.



System User – Any individual who interacts with the system, including administrators, coordinators, and students.

Usability (ISO 25010) – The degree to which the system is easy to use, learn, and understand for its intended users.

User Authentication – The process of validating user credentials before granting access to the system.

User Experience (UX) – The overall satisfaction and ease of use experienced by users when interacting with the system.

User Interface (UI) – The visual elements and layout through which users interact with the system.

Web Application – A software application that runs on a web browser and is accessed via a network.